MetiTrack: Manufacturing Process Tracking & Resource Telemetry Platform

Technical Assignment 1-Page Summary | Candidate: **Khant Lin** (khant20lin061@gmail.com) | Meticuly Co., Ltd. (Ref Code: **150102**)

138/138 Tests 100% Green | **ISO 13485 & FDA 21 CFR Part 11**

1. SUMMARY OF THE REQUIREMENTS

Engineered a regulatory-compliant **Manufacturing Execution System (MES)** tailored to Meticuly's core business: custom 3D-printed Titanium Patient-Specific Implants (PSI) under cleanroom **ISO 13485:2016** and **US FDA 21 CFR Part 11 / Part 820**:

- **Sequential State Transitions:** Trace implant orders chronologically through multi-station cleanroom routing without skipping or reversing stages.
- **Shop-Floor WIP Visibility:** Concurrently triage active cases across each distinct cleanroom station in real time.
- **Consumable Resource Telemetry:** Deduct and compute consumption of Titanium Ti-6Al-4V Grade 23 powder (grams), SLM/CNC machine time (hours), and labor.
- **Executive Dashboard:** Real-time cleanroom bottlenecks, load distribution, and cumulative financial accounting.
- **Traceability & eDHR:** Immutable audit trail satisfying ISO 13485:2016 Clause 7.5.8 and FDA 21 CFR Part 830 Unique Device Identification (UDI).

2. CONVERTING REQUIREMENTS INTO FUNCTIONAL SPECS

- **Strict 5-Station State Machine:** `CAD` → `SLM Print` → `CNC Post-Processing` → `QA Inspection` → `Packaging`. Invalid jumps and backwards regressions are strictly blocked.
- **Atomic Depletion & Concurrency Control:** Advancing an order atomically logs BOM usage and updates balances (`Stock_new = Stock_old - qty`). Pessimistic locking (`select_for_update`) with deterministic ID sorting prevents overselling and deadlocks.
- **Cumulative Cost Engine:** Real-time calculation `Total Cost = ∑(Qty × Unit Cost)` across Materials, Machine, and Labor.
- **FDA Part 11 Asymmetric Signatures:** Electronic DHRs are cryptographically sealed using **ECDSA NIST P-256 (FIPS 186-5)** with SHA-256 fingerprints.
- **HIPAA Privacy:** Patient references encrypted at rest using authenticated **AES-256-GCM with HKDF key derivation** (45 CFR § 164.312).

3. BRIEF DESCRIPTION OF SOFTWARE DESIGN

- **Domain 1: Persistence Layer (PostgreSQL / SQLite):** Normalized relational schema (`ProcessStep`, `Resource`, `Order`, `OrderResourceConsumption`, `OrderStepLog`, `OperatorProfile`, `SecurityAuditEvent`). Append-only immutability.
- **Domain 2: Business Logic & Invariants (Django REST 3.15 / Python 3.13):** Decoupled services (`OrderService`, `InventoryService`, `CostService`, `PKIService`, `PHIEncryptionService`). Zero-Trust JWT, HttpOnly cookies, CWE-1236 CSV injection defense, rate limiting.
- **Domain 3: Cleanroom Presentation Layer (Next.js 16 Turbopack / React 19 / Tailwind 4):** Native typed API service (`lib/api.ts`), horizontal 5-station Kanban, dynamic BOM modals, and real-time streams.

3. SOFTWARE DESIGN (CLEANROOM FEATURES)

- **Interactive Three.js 3D Titanium Viewport:** 480px slide-over drawer with 360° orbital rotation, solid vs. wireframe FEA inspection, and SLM machine specs.
- **Automated FDA UDI Label Generator:** ISO 15223-1 cleanroom packaging label with GS1-128 linear barcode and 2D GS1 DataMatrix (ECC200).
- **Telemetry & Warehouse Views:** Recharts cleanroom WIP load distribution, warehouse stock replenishment modal, and master queue dense table.
- **Real-Time Security Audit Stream:** Immutable ISO 13485 / FDA 21 CFR Part 11 append-only log tracking auth, rate limiting, and cryptographic tamper events.

4. SIMPLIFIED USER MANUAL & CERTIFIED ROLES

Stack URLs: Frontend: `http://localhost:3001` | API & Swagger: `http://localhost:8001/api/docs/`

Certified Personas (Quick Role Switcher):
- **Plant Admin:** `admin` / `MeticulyAdmin2026!` (Full access across all stations)
- **Biomedical Engineer:** `biomed_lead` / `Biomed2026!` (Station 1: CAD Modeling)
- **Production Operator:** `slm_tech` / `Operator2026!` (Stations 2 & 3: SLM & CNC)
- **QA Specialist:** `qa_lead` / `QASpecialist2026!` (Stations 4 & 5: QA & eDHR Sign)

Quick Run: Click **+ New Case** to intake patient → Click **Advance** → on Kanban card → Adjust Titanium powder & machine hours → Click **Advance**. Click any card to inspect 3D WebGL mesh, verify ECDSA signature, and print sterile barrier FDA UDI labels.

5. SUMMARY OF THE SOFTWARE

MetiTrack provides Meticuly with a verified digital spine for medical 3D titanium implant manufacturing. Rigorously validated with **138/138 passing automated tests across 23 test suites** (105 backend Pytest suites + 28 frontend Vitest suites + 5 Playwright browser E2E suites — 100% pass rate), turnkey Docker containerization (`docker compose up --build`), and green GitHub Actions CI/CD. It guarantees data integrity, eliminates procedural errors, and enforces ISO 13485 cleanroom traceability.